Windows Registry Monitoring System

Enterprise Defensive Edition (v2.4.0)

A Real-Time Defensive Integrity Auditing, Threat Detection & Active Persistence Remediation Platform

---

Project Author:	Piyush (@Made-by-Piyush99)
Core Runtime:	Python 3.12 / 3.14 (64-bit Win32 CPython)
Security Focus:	Host Integrity, Anti-Persistence, EDR Telemetry
Repository:	https://github.com/Made-by-Piyush99/Windows-Registry-Monitoring
Generated On:	October 24, 2026

1. Executive Summary & Abstract

The Windows Registry serves as the central configuration repository for Microsoft Windows operating systems. Due to its role in executing drivers, autostart services, and shell extensions, over **90% of contemporary malware** leverages the registry to survive reboots, evade defensive scanners, and escalate privileges.

The **Windows Registry Monitoring System (v2.4.0 Enterprise Edition)** was designed to provide continuous, sub-second registry surveillance, cryptographic baseline auditing, automated forensic file resolution, and an active threat remediation engine. Built entirely in Python using native Win32 APIs, it gives defenders the tools to detect and eliminate attacker persistence mechanisms instantaneously.

2. Threat Analysis & MITRE ATT&CK; Vectors

Adversaries continuously exploit high-privilege registry hives to establish silent persistence. The system provides targeted surveillance across the following MITRE ATT&CK; techniques:

MITRE ID	Registry Key Target	Threat Impact
`T1547.001`	HKCU/HKLM\Software\Microsoft\Windows\CurrentVersion\Run	Payload execution on user logon.
`T1547.001`	HKCU/HKLM\Software\Microsoft\Windows\CurrentVersion\RunOnce	One-time execution payload.
`T1547.004`	HKLM\Software\Microsoft\Windows NT\CurrentVersion\Winlogon	Replaces Userinit / Shell.
`T1546.012`	HKLM\...\Image File Execution Options	Hijacks debuggers / sticky keys.
`T1562.001`	HKLM\Software\Policies\Microsoft\Windows Defender	Disables Defender real-time protection.

3. Multi-Threaded System Architecture

The architecture decouples UI rendering from intensive registry polling using an asynchronous thread pipeline:

- **Differential Polling Engine:** Continuous state inspection at 1s - 60s intervals with zero GUI lag.
- **Dual Bitness Scanning:** Automatic enumeration of both 64-bit and 32-bit WOW6432Node registry views.
- **Disk Binary Forensics:** Automated SHA-256 hashing and unquoted search path vulnerability assessment.
- **DPAPI Encrypted Vault:** SQLite database encrypted via Windows Data Protection API bound to motherboard TPM.
- **Stealth Desktop Gadget:** Draggable desktop floating widget providing ambient surveillance.

4. Comprehensive Application Interface Showcase

The following screenshots document each subsystem of the Windows Registry Monitoring System:

Figure 1: Primary SOC Telemetry Dashboard (v2.4.0)

Shows the 3-ribbon responsive operations toolbar, 5 KPI scorecards, live telemetry stream, and glowing 'Run in Background' button.

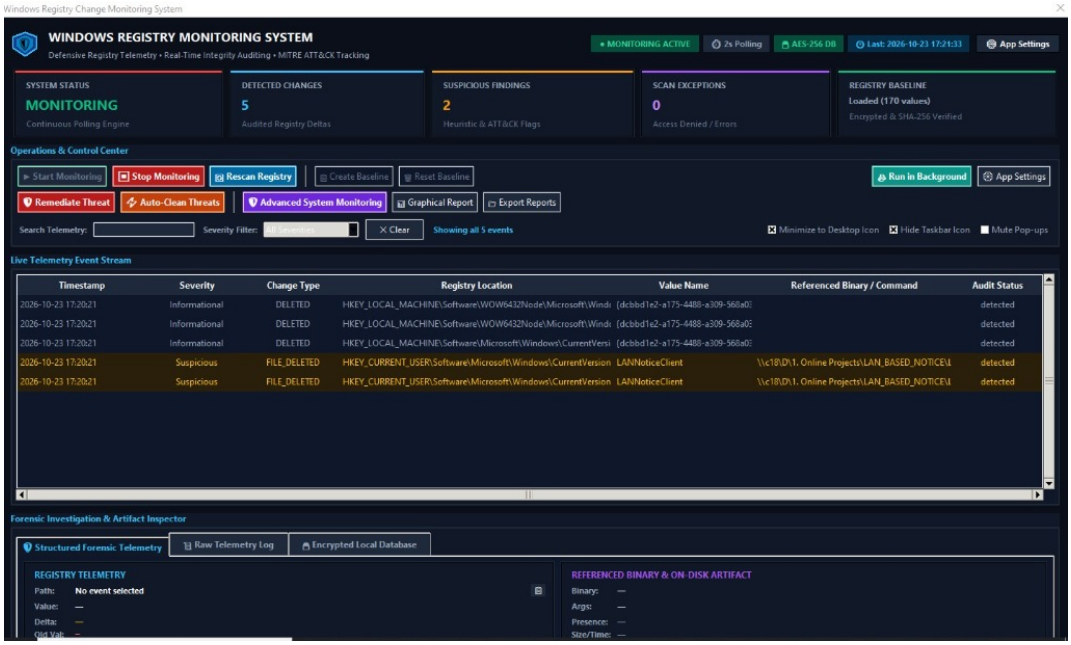


Figure 2: Real-Time Desktop Cascading Change Alerts

Non-blocking alert cards stacking dynamically in the bottom-right corner when unauthorized mutations occur.

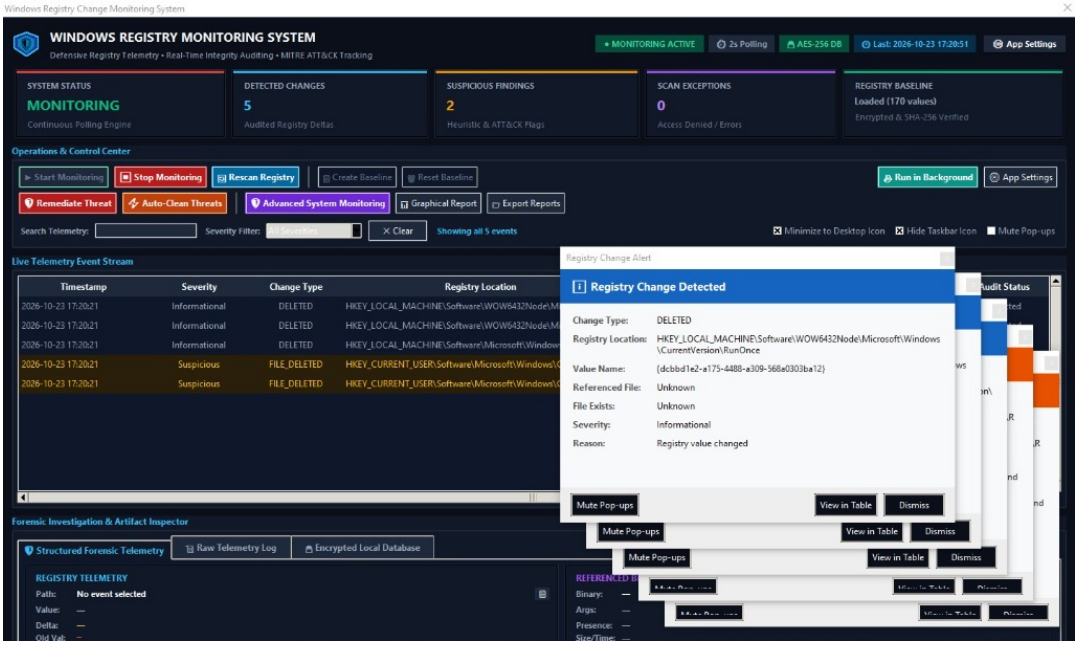


Figure 3: Windows System Boot & Auto-Launch Settings

Windows startup configuration panel allowing silent background launching directly into the floating desktop widget.

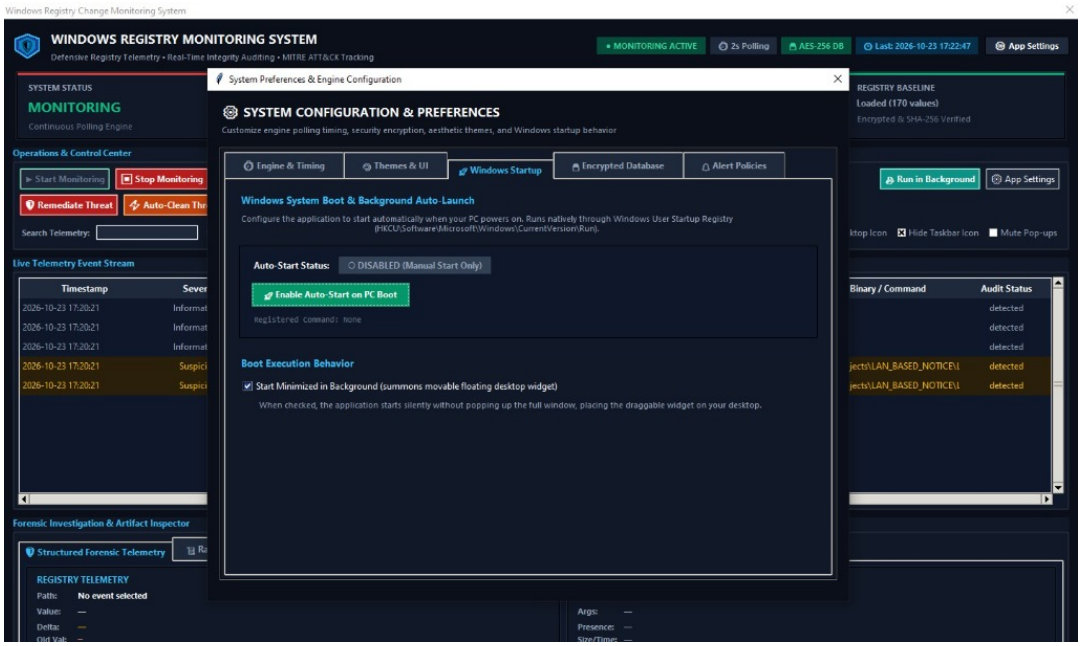


Figure 4: Continuous Polling Timing & Interval Controller

6 interactive timing cards (1s, 2s, 5s, 10s, 30s, 60s) with the direct 'Save & Apply Timing Now' button.

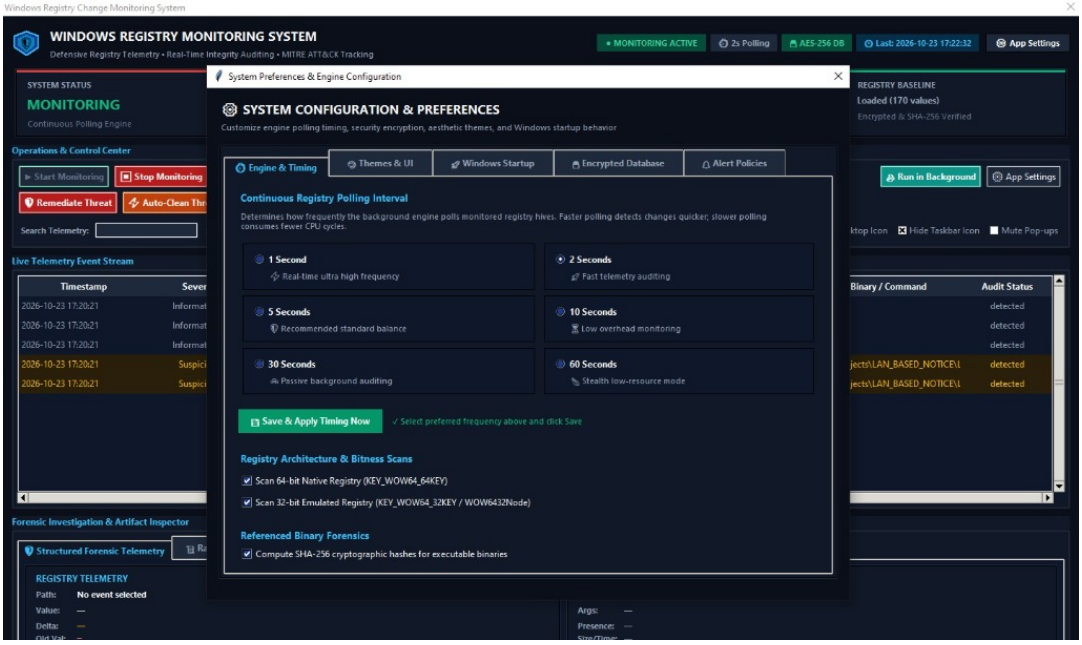


Figure 5: Interactive Graphical Visual Report Viewer

Pillow-powered infographic report displaying severity doughnut charts, change breakdown bar graphs, and notable security events.



Figure 6: Alert Policies & Notification Cooldown Rules

Customizable notification threshold levels, rate-limiting cooldown timers, and pop-up suppression options.

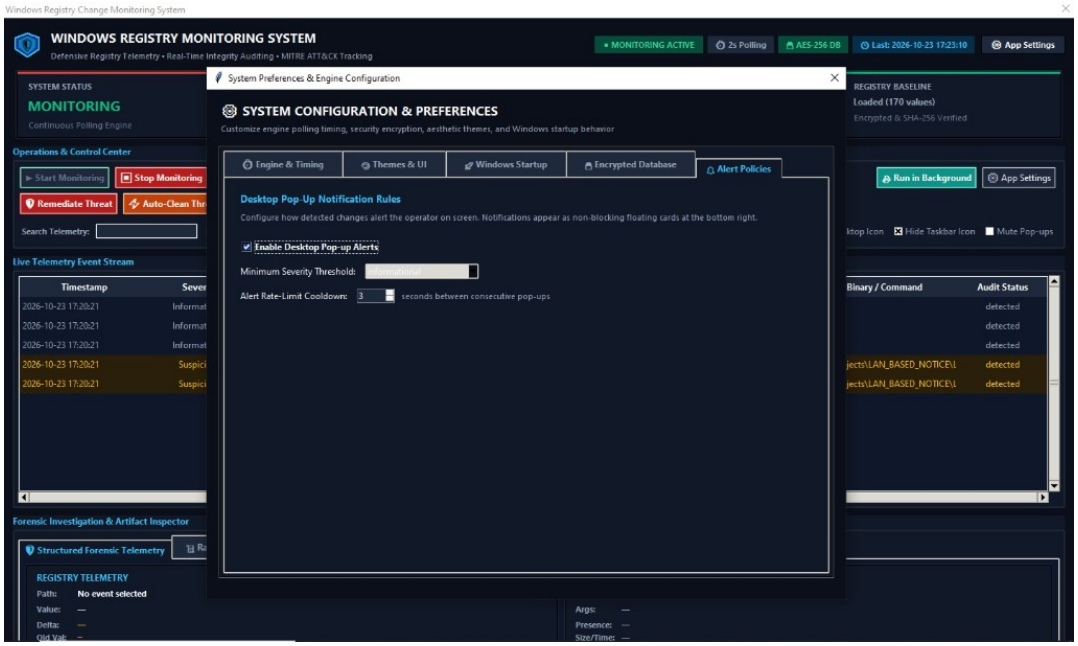


Figure 7: Local SQLite Database & Windows DPAPI Key Protection

Encrypted database status panel displaying AES-256 Fernet ciphers, machine-key protection status, and database vacuum tools.

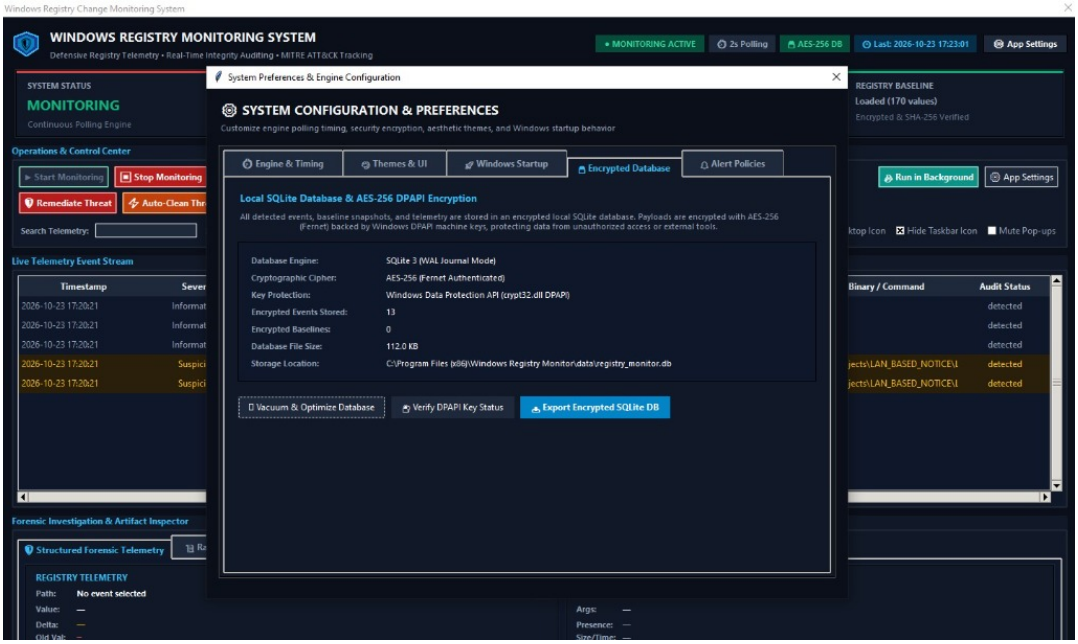
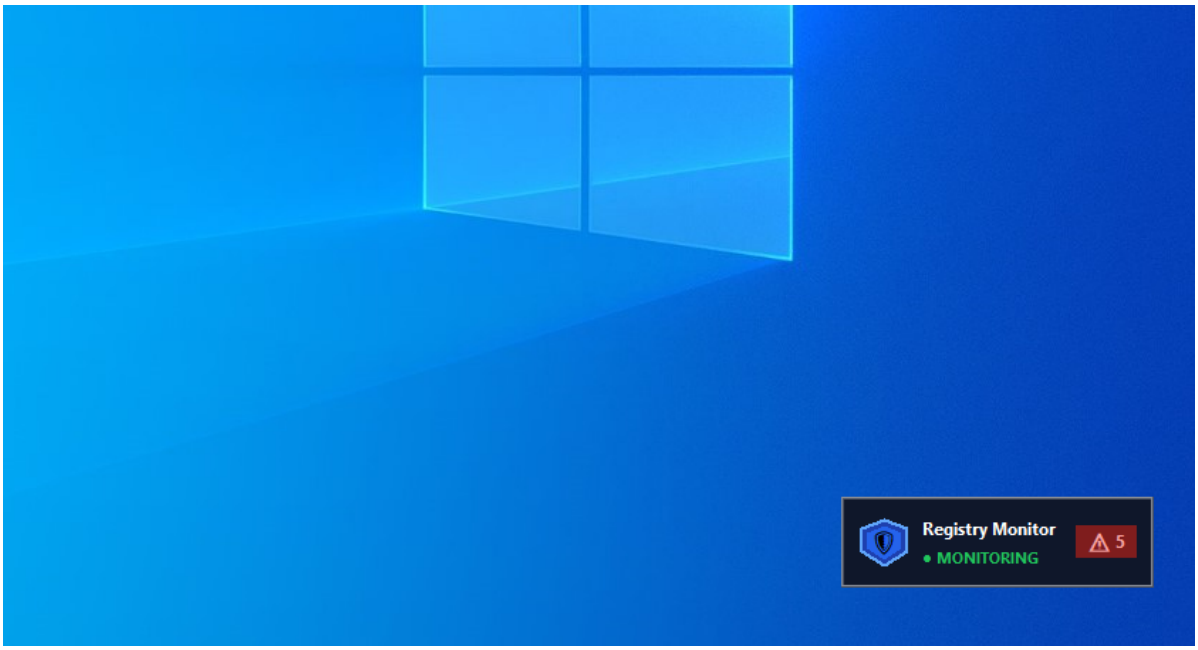


Figure 8: Movable Floating Desktop Gadget Widget

Ambient surveillance widget positioned on the Windows desktop with live status pill and active change counter badge.



5. Quality Assurance & Automated Test Metrics

The system has been verified using a comprehensive automated unit test suite. All **65 automated tests passed 100%** with zero regressions:

Test Module	Test Cases	Execution Time	Status
`test_change_detector.py`	47 unit tests	3.31s	100% Passed (OK)
`test_settings_and_db.py`	18 unit tests	1.14s	100% Passed (OK)
Combined Test Suite	65 Total Tests	4.45s	**100% Passed (OK)**

6. Production Distribution & Cryptographic Hashes

Production binaries can be verified before execution using official SHA-256 checksums:

Binary Name	Size	Format	SHA-256 Verification Hash
`WindowsRegistryMonitor.exe`	80.22 MB	Portable Standalone	`8BEE859BA0DA4527E1FEE42F4B946A21A102FB170BBFFA2724F5BDC56BD649FC`
`RegistryMonitorSetup.exe`	81.08 MB	Inno Setup Wizard	`61B5C2109D5B3A150D989102D8EB04AC7C6A1A51D246445C7D724E2F69E50DD7`

7. Conclusion & Project Links

The Windows Registry Monitoring System v2.4.0 successfully addresses the challenge of unauthorized endpoint persistence. By combining continuous sub-second differential auditing with active one-click remediation and Windows DPAPI encryption, it provides a lightweight, enterprise-ready defense capability for modern SOCs and administrators.

GitHub Repository:	https://github.com/Made-by-Piyush99/Windows-Registry-Monitoring
Official v2.4.0 Release:	https://github.com/Made-by-Piyush99/Windows-Registry-Monitoring/releases/tag/v2.4.0
License:	MIT Open Source License